

Predicting The

Future Stock Volatility

with Interpretable Models

Presentation for Louis, Quantitative Trader

I.

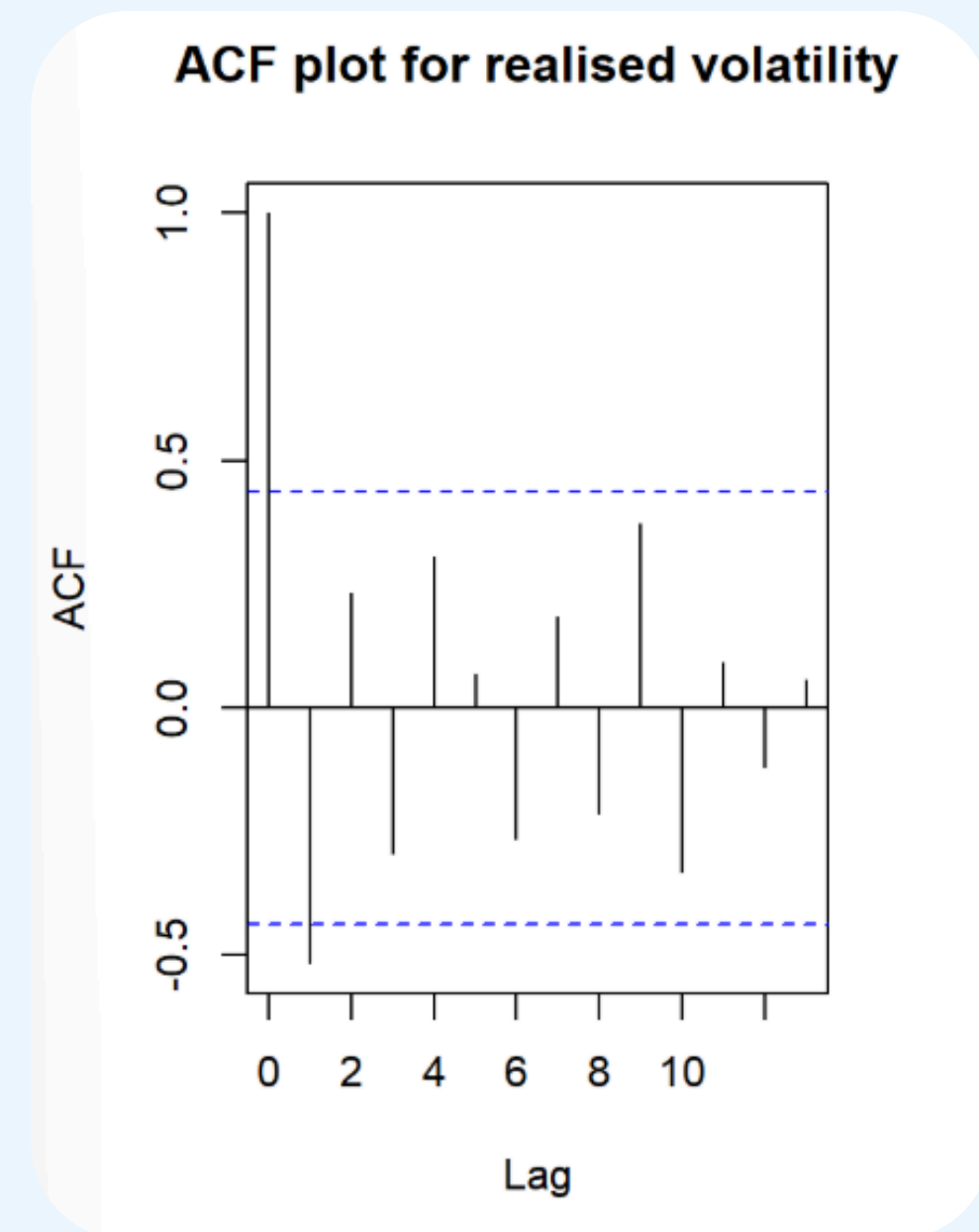
Goal

Build interpretable model to
predict future volatility

II.

Data

stock_1 data
1,507,532 rows & 11 columns

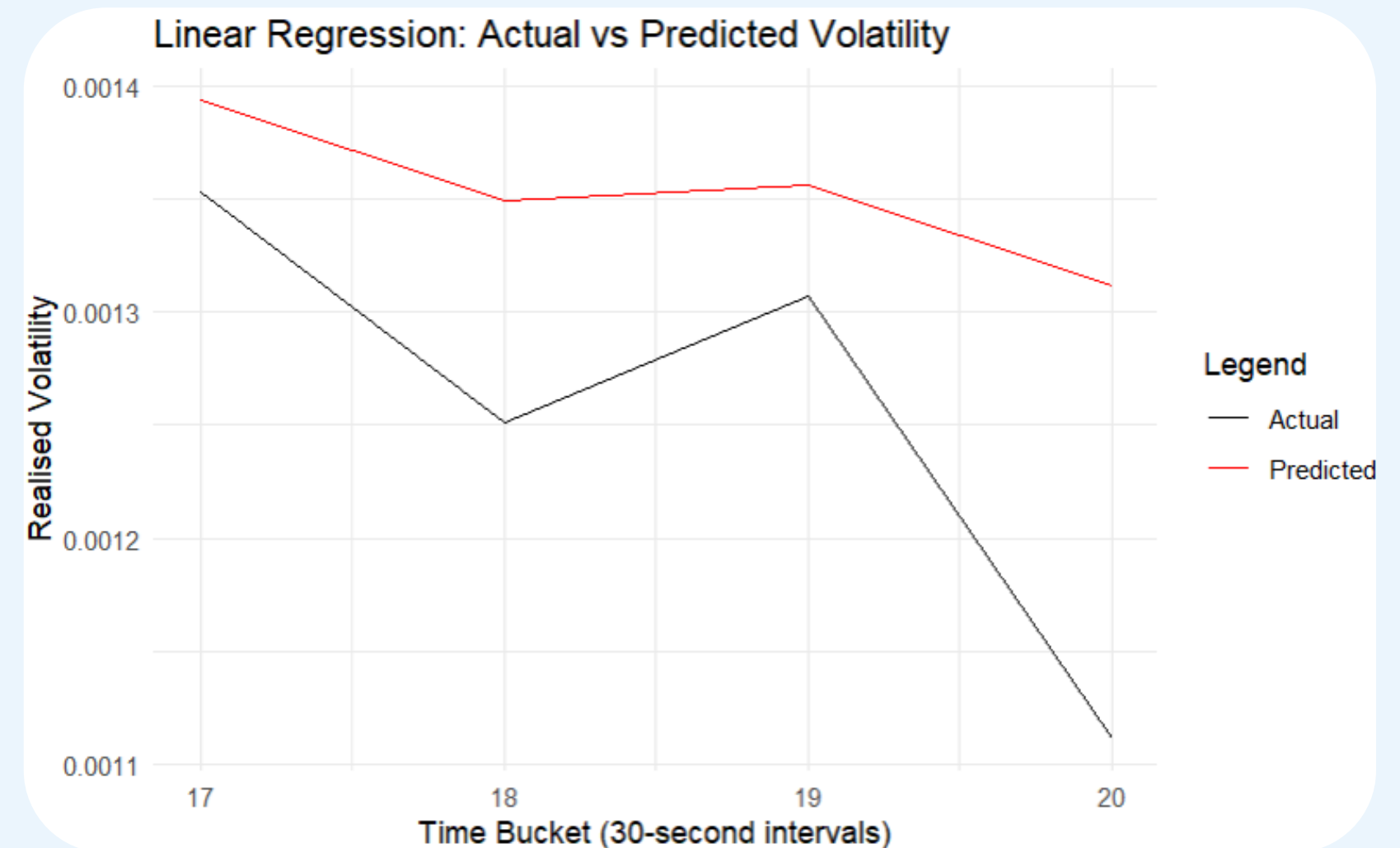


data subset to 500 time intervals
representing a 10-minute period

I. Linear Regression

Baseline & simple model using predictors:
price, order volume, bid-ask spread

Build weighted regression on training data
(80%) → split into 30-second intervals



In-sample performance
 $R^2 \approx 74.1\%$

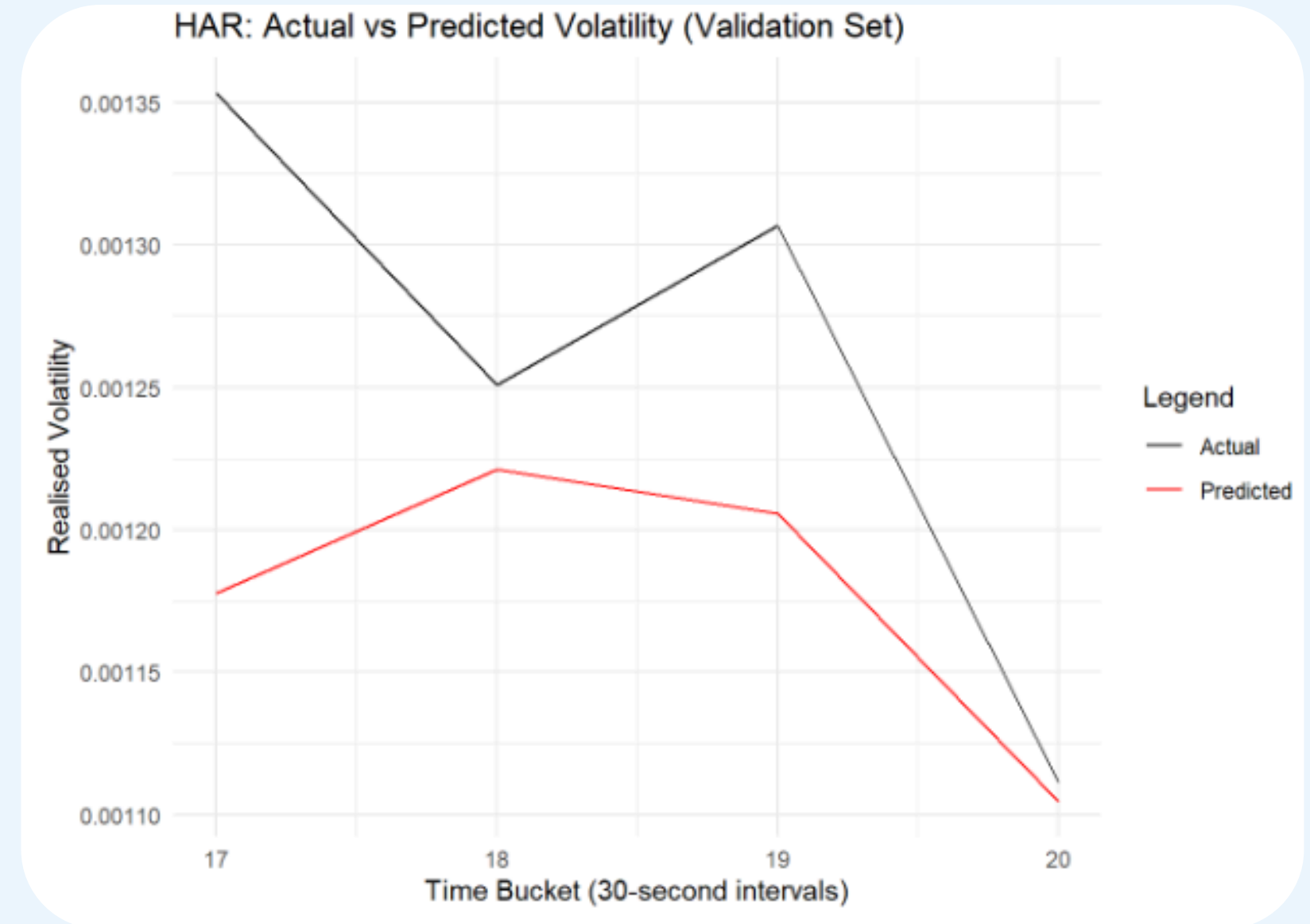
II. HAR

Out-of-Sample (Validation)

Assess how well the model generalises

Use historical volatility (lagged and averaged) to capture short to long-term

Improve predictions → past volatility and averages over different time periods



In-sample performance

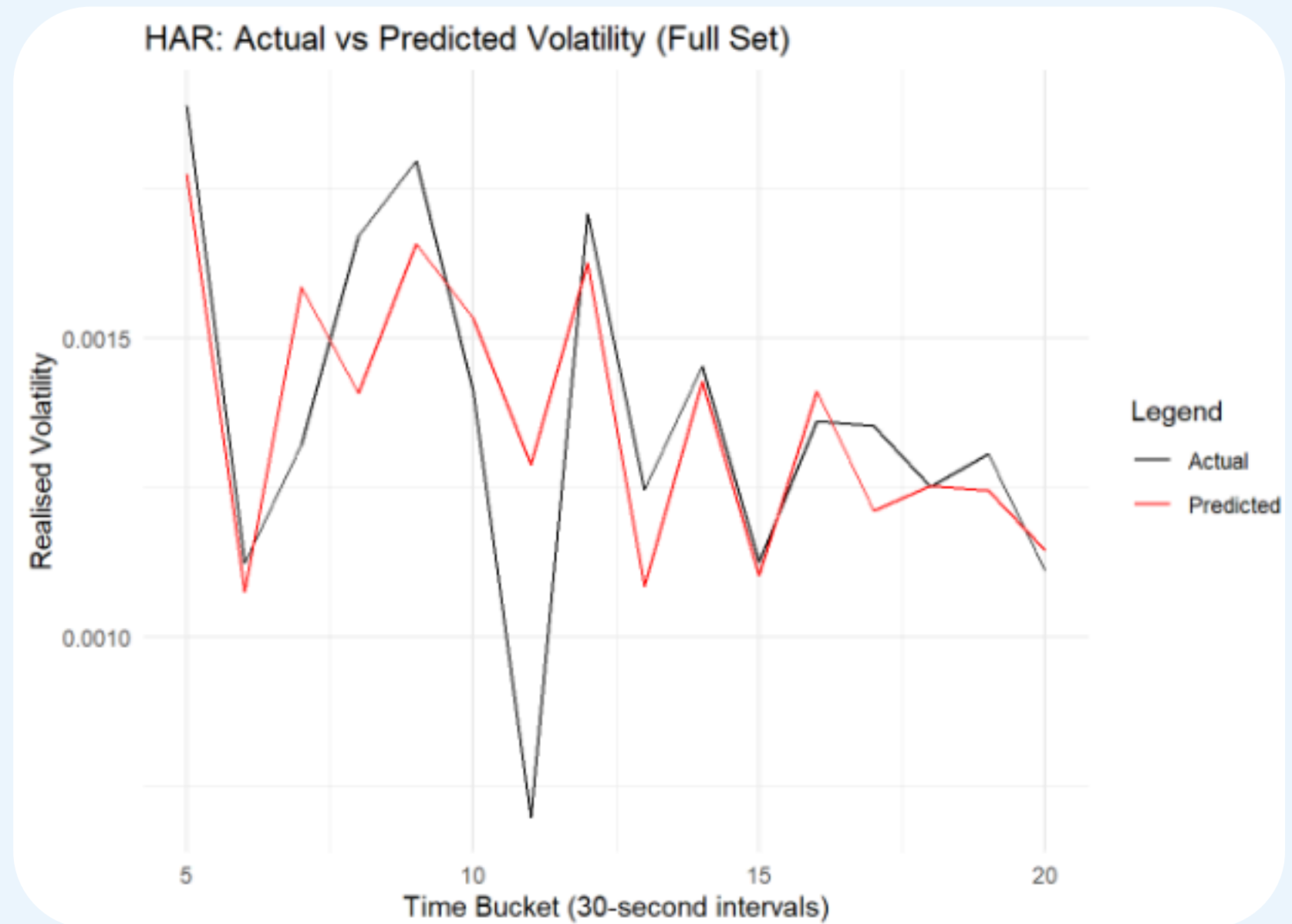
$R^2 \approx 54.3\%$

II. HAR

In-Sample (Full Set)

Understand overall fit

Combines short-term and longer-term past volatility information to smooth out prediction errors

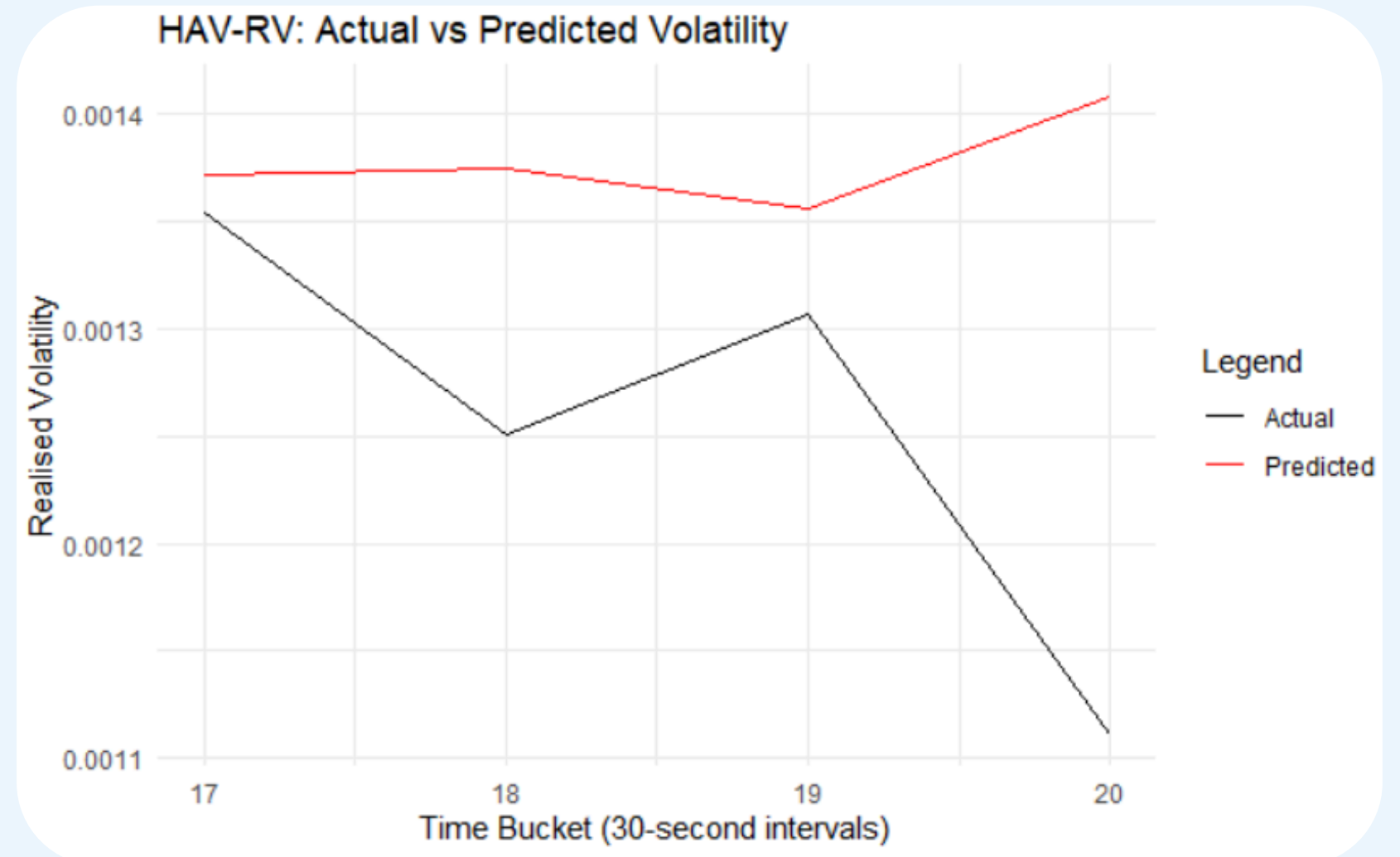


In-sample performance
 $R^2 \approx 55.3\%$

III. HAV-RV

Use both past values and averages over past 5 intervals

Apply weighted version → adjust noisy & jumpy data



In-sample performance
 $R^2 \approx 85.6\%$

Evaluation Framework

RMSE

Measures overall prediction error
(lower is better)

MAPE

Gives Average percentage error
(lower is better)

R^2

Indicates how much variance is
explained by the model
(closer to 1 is better)

MAE

Gives the Absolute error magnitude
(lower is better)

Validation Strategy

- 80% training, 20% validation split across intervals
- Out-of-sample validation to test generalisability

Model Evaluation

Model	RMSE	MAE	R ²	MAPE
Linear Regression	0.000116	0.000097	-0.626035	8.15%
HAR (Out-of-Sample)	0.000102	0.000078	-0.269537	5.91%
HAR (In-Sample)	0.000193	0.000132	0.552935	11.91%
HAV-RV	0.000287	0.000287	0.000287	28.71%
GARCH	0.001078	0.00108	-17,629	85.03%



Final Model Selection

HAR (Out-of-Sample) Model

Lowest RMSE, MAE, and MAPE

- Best overall error metrics & proving its reliability for unseen data
- Incorporates lagged volatility and averages, allowing it to capture realistic trends

Trade-offs & Limitations

- Negative R^2 of the HAR validation
- High-frequency noise

Future Considerations

- Scale up/expand analysis
- Explore additional predictors
- Further testing over different periods

Closing The Loop

Final Thought

Additional Exploration

Beyond What Were Presented

Data Processing

Weighted Average Price (WAP)

bid and ask prices and sizes

measure of average price of stock

Log>Returns

basis for deriving volatility

measure of percentage change over time

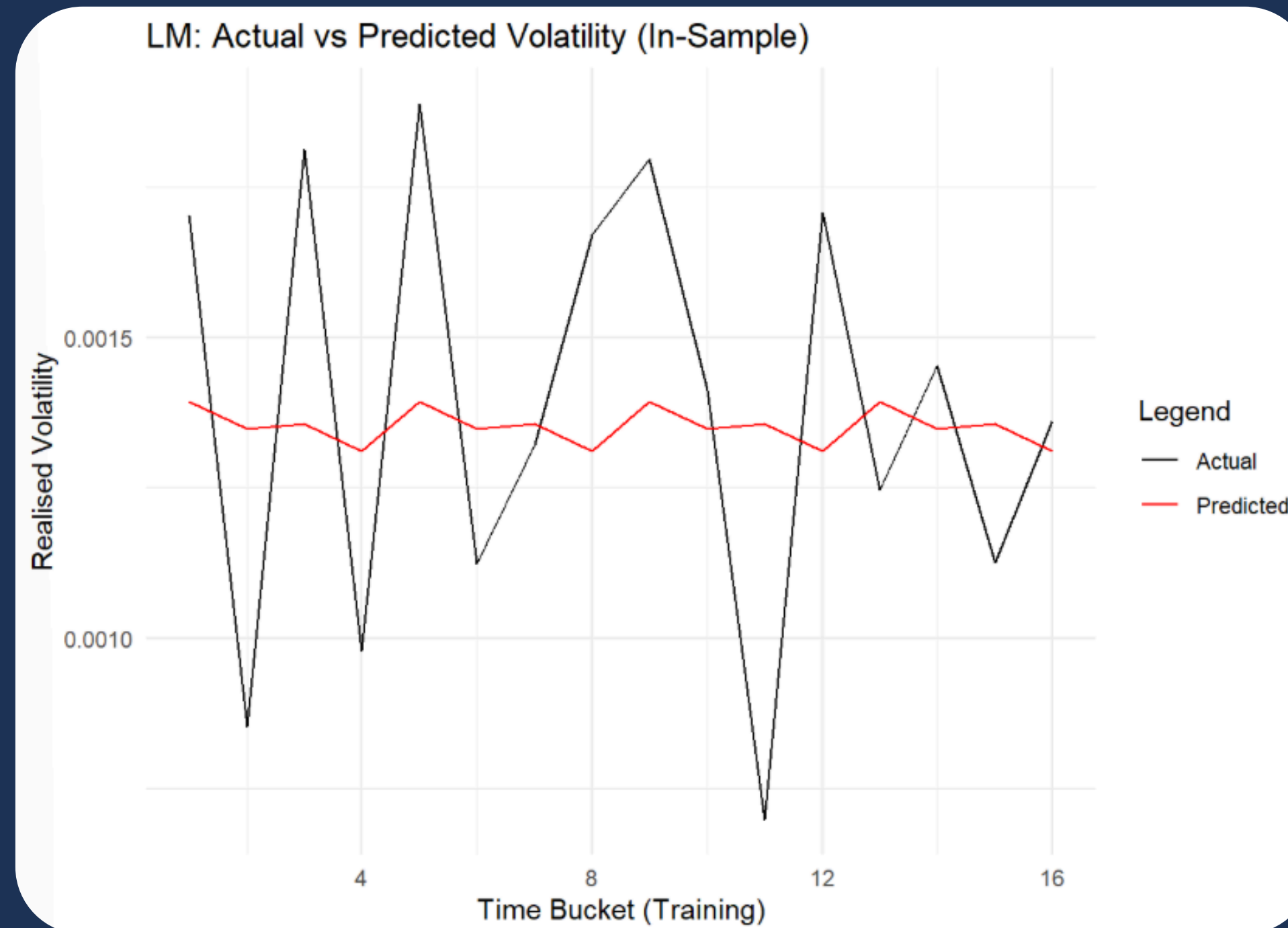
Aggregation

reliased volatility → aggregated into 30-second buckets

measure of actual price fluctuations over time

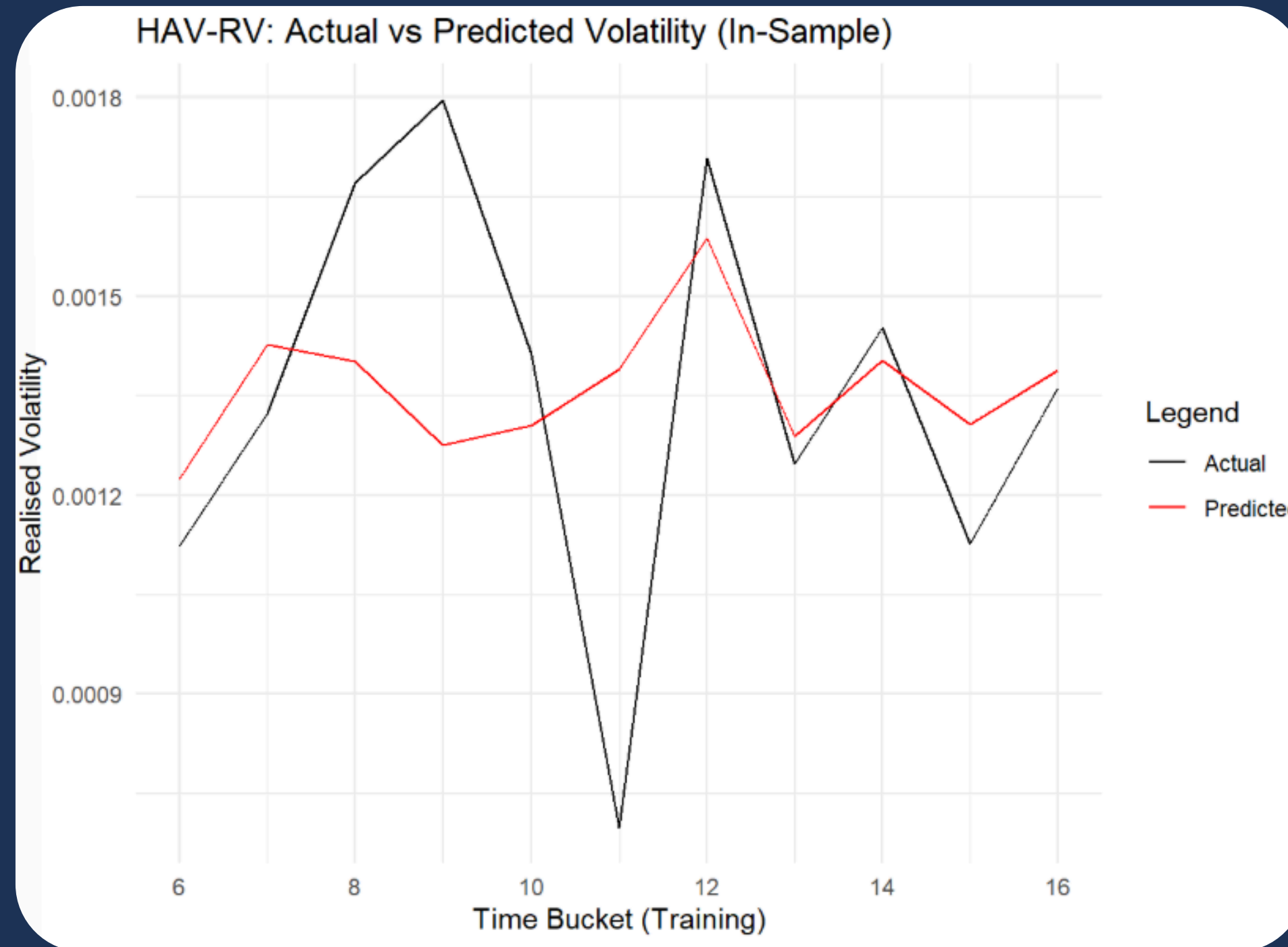
Linear Regression

In-Sample



HAV-RV

In-Sample



GARCH

In-Sample

